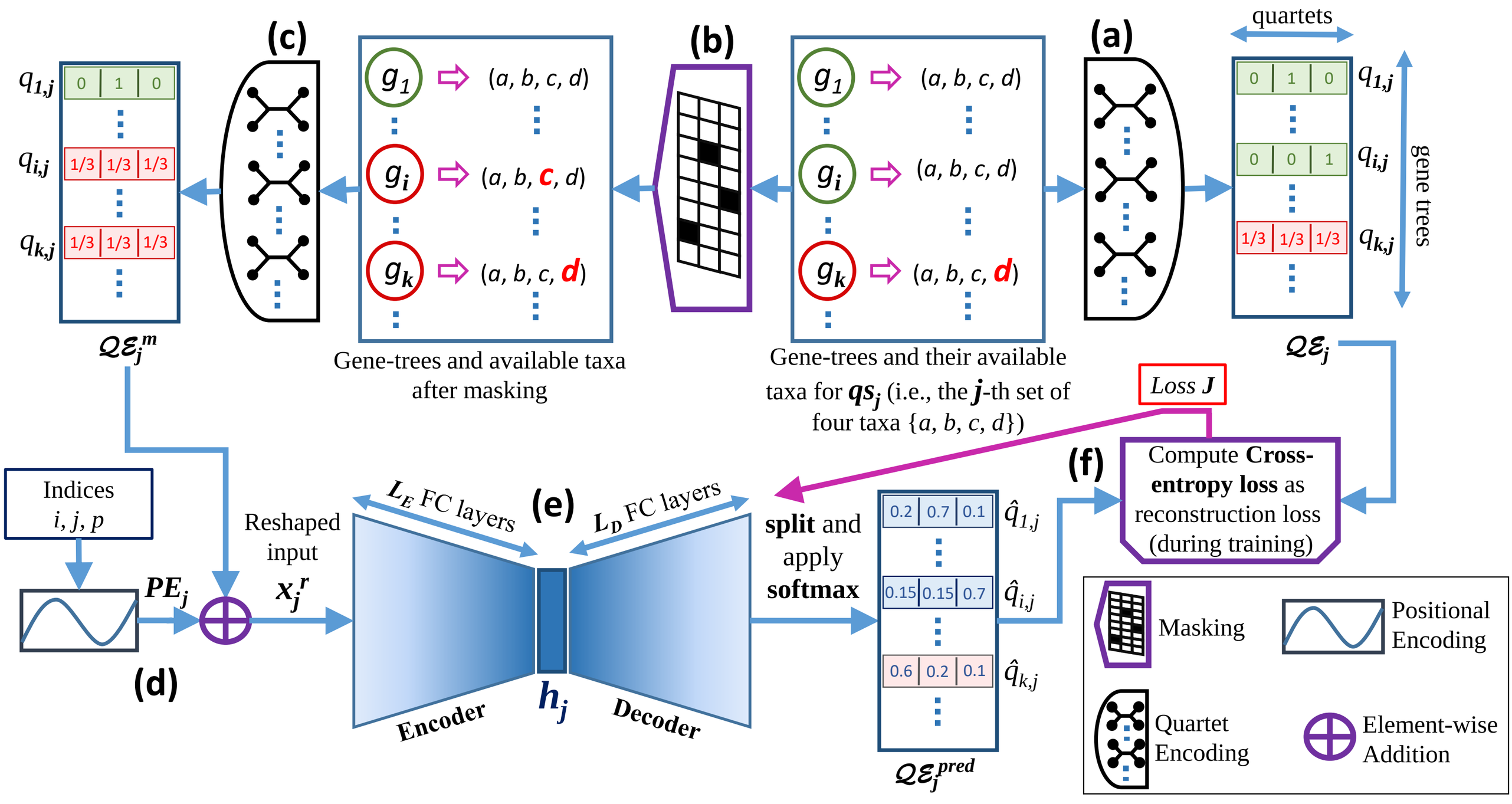

(a)
quartets
gene trees
$q_{1,j}$
$q_{i,j}$
$q_{k,j}$
0 | 1 | 0
0 | 0 | 1
1/3 | 1/3 | 1/3
$\mathcal{QE}_j$
(b)
g_1 ⇨ (a, b, c, d)
g_i ⇨ (a, b, c, d)
g_k ⇨ (a, b, c, d)
Gene-trees and their available taxa for qs_j (i.e., the j-th set of four taxa {a, b, c, d})
(c)
Gene-trees and available taxa after masking
$\mathcal{QE}_j^m$
Loss J
(f)
Compute Cross-entropy loss as reconstruction loss (during training)
split and apply softmax
0.2 | 0.7 | 0.1
0.15 | 0.15 | 0.7
0.6 | 0.2 | 0.1
$\hat{q}_{1,j}$
$\hat{q}_{i,j}$
$\hat{q}_{k,j}$
$\mathcal{QE}_j^{pred}$
(e)
L_E FC layers
L_D FC layers
Encoder
h_j
Decoder
Reshaped input
x_j^r
Indices
i, j, p
PE_j
(d)
Masking
Positional Encoding
Quartet Encoding
Element-wise Addition